

Randomized and low-rank approximation

Optimal dimension for a rerandomized Hadamard embedding

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Rating rationale: Challenging because two structured randomizations must remove the embedding logarithm uniformly over subspaces; community impact is fast sketching for least squares and low-rank methods.

LEAN VERIFIED — complete affirmative resolution. TR-01 is resolved by the v7 manuscript theorem with full two-randomization rerandomized SRHT statement at prescribed width. The result is not a weaker-width selection. This retained entry is excluded from the open count.

Problem statement

Let n be a power of two, $1 \leq r \leq n$, and $0 < \varepsilon < 1/2$. Let F be the normalized Walsh–Hadamard matrix, let D_1, D_2 have independent Rademacher diagonal entries, and let $S \in \mathbb{R}^{n \times k}$ select a uniformly random k -element subset of coordinates, independently. Set

$$\Omega = \sqrt{n/k} D_1 F D_2 F S.$$

Does a universal $C > 0$ exist such that, for every fixed r -dimensional subspace $V \subseteq \mathbb{R}^n$, choosing $k = \min\{n, \lceil Cr/\varepsilon^2 \rceil\}$ gives

$$\Pr \{ (1 - \varepsilon) \|x\|_2^2 \leq \|\Omega^T x\|_2^2 \leq (1 + \varepsilon) \|x\|_2^2 \text{ for every } x \in V \} \geq 0.99?$$

This fixes normalization, sampling without replacement, and a constant success probability in the workshop question. Results for a single randomization, three randomizations, or independently sampled sparse sketches do not establish this statement.

References

Amsel et al., *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, §5.3, Definition 5.3 and Problem 5.6. For the single-round baseline, Joel A. Tropp, *Improved Analysis of the Subsampled Randomized Hadamard Transform*, *Advances in Adaptive Data Analysis* 3 (2011), 115–126, Theorem 3.1.

Resolution and status check — 2026-09-12

The manuscript *Subspace embeddings with the rerandomized SRHT* (Theorem 1, manuscript page 2, [manuscript PDF](#)) proves that for each power-of-two n , every $1 \leq r \leq n$, and every fixed r -dimensional subspace V , choosing

$$k = \min \{n, \lceil Cr/\varepsilon^2 \rceil\}$$

gives

$$\Pr \{ \|V^T \Omega \Omega^T V - I_r\|_2 > \varepsilon \} \leq 0.01,$$

for two independent Rademacher sign diagonals, two normalized Walsh transforms, and a uniform k -subset coordinate sample without replacement. The theorem proves the *prescribed width itself* with full

width cap and does not rely on a smaller alternative k . The same universal constant works for every power-of-two n , and there are no remaining parameter cases.

The proven range is $0 < \varepsilon < 1$, which strictly contains TR-01's displayed range $0 < \varepsilon < 1/2$. This is therefore a complete affirmative resolution of the original target.

TR-01 is a prescribed-width strengthening of Problem 5.6 in the 2026 Simons workshop collection. It uses the same two-round Walsh/sign distribution and the same fixed-subspace OSE event, but TR-01 prescribes the specific width $k = \min\{n, \lceil Cr/\varepsilon^2 \rceil\}$ rather than only asking for a width of order $O(r/\varepsilon^2)$. The manuscript proves this stronger prescribed-width formulation directly.

Lean proof and verification evidence

The public Lean repository is [OpenProblemsInNLA_TR-01@ed211811](#), and the v7 Lean entry point is [lean/Problem56/PaperV7/Certification.lean](#).

The canonical result declarations are

- [Problem56.PaperV7.certified_main](#)
- [Problem56.PaperV7.certified_explicit_main](#)
- [Problem56.PaperV7.main_prescribed_width_ose](#)

The submitted manuscript and correspondence evidence are available at

- [manuscript.pdf](#)
- [manuscript.tex](#)
- [final_candidate_correspondence.md](#)
- [final_candidate_correspondence.json](#)

Lean toolchain and dependency pins are

- [lean-toolchain](#) ([lean 4.33.0](#))
- [lake-manifest.json](#) (mathlib [db584cd6d46c92f209a44c0f1c829460d327499d](#))

The pinned reproduction instructions are in [lean/README.md](#), and the verifier is [lean/verify.py](#). From the [lean/](#) directory of that immutable revision, the principal commands are:

- `lake exe cache get`
- `python3 verify.py`
- `python3 verify.py --graph --semantic --fresh-kernel`

The archived public record reports successful proof compilation, statement-boundary checks, transitive axiom checks, semantic regressions, a cold project rebuild, and fresh kernel replay.

Reproducible checks are recorded in the immutable validation artifacts:

- [delivery_check.json](#) (PASS)
- [cold_rebuild.json](#) (rebuild log and source hashes)
- [fresh_kernel_replay.json](#)
- [semantic_regressions.json](#)
- [final_binding_review.json](#)

The axiom audit is in the immutable artifact [AxiomAudit.lean](#), with accepted transitive axioms only

- [propext](#)
- [Classical.choice](#)
- [Quot.sound](#)

No `sorry`, `admit`, or custom mathematical axioms appear in the certified target graph.

The submitted verification package includes an independent manuscript-to-Lean statement-correspondence review and the public verification records linked above. This pull request does not claim that the OpenProblemsInNLA maintainers reran Lean locally; the formal-verification status is supported by the immutable public proof and audit record submitted for maintainer review.